

Portfolio of Evidence, Part 1 – Task 1 (Research)

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1. Introduction

The Smart-X gateway ingests multi typed telemetry from thousands of ESP32 nodes. At that volume the binding constraint on the operator is not data availability, it is attention. A console that presents every reading with equal visual weight forces a serial search for the one node that has failed. This report identifies five engagement strategies drawn from the visualisation, human factors and information systems literature, then justifies the one selected for implementation.

2. Candidate Dashboard Engagement Strategies

2.1. Pre-attentive anomaly encoding

Severity is carried by low level visual channels, hue, motion and luminance, rather than by text, so an outlier is located in a single fixation regardless of how many nodes are on screen (Healey & Enns, 2012). This is the strategy selected below.

2.2. Embedded sparkline micro history

Each device tile carries a compact trend line of its recent packets, so the operator sees not only the current value but its trajectory. Trend context is what distinguishes a transient sensor glitch from genuine gradual drift, and it removes the need to open a separate chart to find out.

2.3. Severity tiered, deduplicated alerting

Notifications are ranked by impact and collapsed when the same node repeats the same fault, since acceptance of decision support alerts falls sharply as volume and repetition rise (Ancker et al., 2017). Only the top tier interrupts the operator.

2.4. Progressive disclosure with guided onboarding

The landing view exposes only mesh level state; packet payloads, validation traces and routing detail are revealed on demand. Layering the interface against Endsley's (1995) perception, comprehension and projection levels keep a first-time engineer oriented without hiding depth from an expert.

2.5. Gamified operational health scoring

A composite mesh health score, resolution streaks and per zone leaderboards convert routine maintenance into visible progress. Points, feedback and comparison are the affordances most consistently associated with sustained engagement in gamified systems, though effects are context dependent (Koivisto & Hamari, 2019).



Figure 1: The Sensor Health Pulse Grid.

Chosen Strategy: Justification and Planning

The strategy selected for the Smart-X configuration dashboard is pre-attentive anomaly encoding, implemented as a Sensor Health Pulse Grid. Rather than presenting telemetry as a wall of numeric readouts or a single scrolling log, the dashboard renders one tile per registered node. Three visual channels carry meaning at once. Hue encodes how far the latest reading deviates from the node's own rolling baseline, a brief ring pulse fires on each packet arrival, and a tile desaturates and begins a visible count when its heartbeat lapses.

The justification rests on the distinction between pre-attentive and attentive visual processing. Healey and Enns (2012) show that features such as hue, motion and luminance are resolved by the low-level visual system in roughly 200 milliseconds, and, critically, that detection time for these features is largely independent of the number of distractors present. A numeric table has no such property, locating an outlier among five hundred printed values requires serial search, so troubleshooting time grows with mesh size. Encoding severity as hue makes one red tile among two hundred green ones equally fast to find whether the mesh holds fifty nodes or five thousand. For a system designed to scale thousands of microcontrollers, that difference in scaling behaviour is the entire argument.

Two design decisions follow from the research and target the two failure modes named in the brief. Anomalous spikes are scored per node rather than against a global threshold, using a rolling mean and standard deviation maintained for each device. A greenhouse humidity sensor and a smart meter therefore raise alarms on the same normalised scale, and a reading is flagged because it is abnormal for that device, not merely because it is large. Sensor disconnects are handled by the inverse cue, the absence of data is itself rendered. A silent node fades and displays elapsed silence, so a dropped ESP32 grows more visually distinct over time instead of quietly vanishing from a chart.

Deliberately excluded is a high frequency modal alerting. Ancker et al. (2017) found that clinicians' acceptance of decision support alerts fell sharply as alert volume and work complexity rose, with repeated alerts for the same subject a significant predictor of override. A gateway ingesting thousands of packets per second would generate exactly that pattern, and operators would learn to dismiss the interface. Persistent ambient encoding avoids the trap, state is always visible but never demands acknowledgement, so severity information survives instead of being trained away.

The grid also supports progressive disclosure. It answers Endsley's (1995) Level 1 question, perception of what is happening, at a glance, while clicking a tile expands the sparkline, recent packets and the node's position in the Facility – Zone – SubZone hierarchy, supporting Level 2 comprehension and level 3 projection. Diagnosis therefore begins with orientation rather than with a query. Finally, following

Stelea et al. (2025), each tile carries an ARIA live region text equivalent, so the same state change reaches screen reader users and operators glancing away from the display.

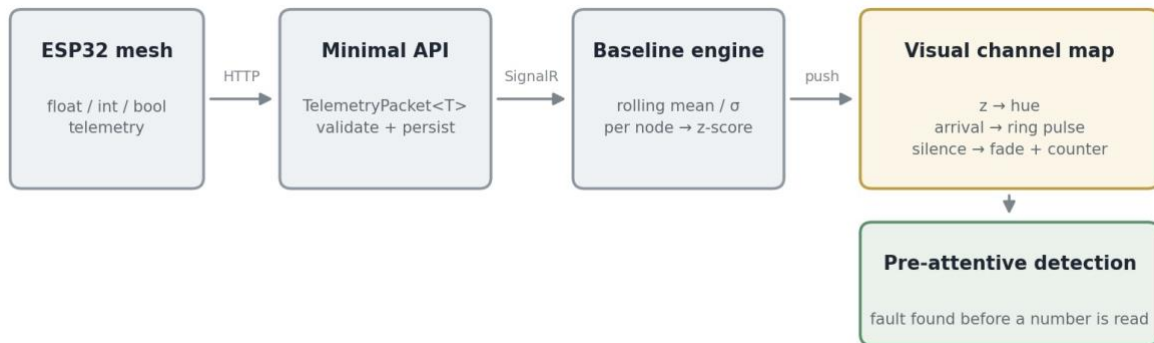


Figure 2: Pipeline from raw packet to visual cue

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